Exemplu examen SO/SO1, 2025/2026

Doar un răspuns este corect la fiecare dintre întrebările tip grilă de mai jos. O întrebare cu răspunsul dat corect și complet primește 10 puncte.
Nota finală se calculează cu formula N=1+(30+r)/20, în care r este numărul de întrebări la care s-a răspuns corect. Observați faptul că așa-numitul "punct din oficiu" e inclus.
Subiectele care nu au răspunsuri tip grilă cer un răspuns scurt, în câteva fraze și pot avea punctaje diferite. Astfel de subiecte pot lipsi la unele prezentări, sau la altele pot fi mai multe. Probabilitatea de a exista mai multe astfel de subiecte crește cu numărul prezentării.
Citiți cu atenție. Nu confundați conceptele între ele. Exemplu: comandă != funcție != proces
Timp de lucru necesar: 20-25 minute. Timp de lucru acordat: 35 minute.

1. Explicati care este diferenta, ca efect, dintre urmatoarele linii de comanda:

```
$  ls -l && tail abc.txt && test -f fis1.c
$  ls -l || tail abc.txt || test -f fis1.c
```

2. Descrieti, in cateva fraze, diferentele si asemanarile de principiu intre pipe-uri si socket-uri.

3. Un mutex este o primitiva care
a) modifica variabile mutabile intr-un thread; b) poate fi utilizata pentru a realiza excluziune reciproca intre thread-uri; c) opreste complet thread-uri; d) partajeaza memoria intre procese; e) se gaseste doar in UNIX si se foloseste pentru a crea un context de multithreading

4. Informatiile despre drepturile de acces la un fisier se gasesc:
a) in nodul index asociat acestuia; b) in fisier; c) in directorul care contine fisierul; d) in sistemul de operare; e) in legatura simbolica

5. Functia kill()
a) termina intotdeauna un proces, in mod brusc; b) trimite un semnal; c) distruge fara drept de apel un proces; d) capteaza un semnal; e) trimite semnalul cu numarul 9 catre comanda curenta

6. Functia alarm()
a) programeaza trimiterea unui semnal anume in viitor; b) semnaleaza o situatie de eroare aparuta la executie; c) reseteaza o alarma declansata; d) informeaza un proces despre aparitia unei erori in alt proces; e) anunta procesele ca trebuie sa se termine imediat

7. Legaturile fixe sunt:
a) intrari in director diferite care refera acelasi nod index; b) nume alternative pentru acelasi fisier, aflate intotdeauna intr-un acelasi director; c) fisiere individuale care refera alte fisiere; d) fisiere individuale care refera directoare fixe; e) fisiere speciale

8. Functia fork() creeaza:
a) o comanda diferita de programul parinte, dar initial identica cu acesta; b) un fiu al programului curent incarcat de pe disc; c) un nou proces cu acelasi identificator ca procesul curent; d) un proces nou; e) un proces parinte

9. Un programator doreste sa transmita un sir de dimensiune variabila mai mica decat 100, prin intermediul unui pipe. Sirul este continut intr-o variabila char s[100] si a fost initializat apeland strcpy(), iar descriptorii asociati pipe-ului se afla in variabila pfd. Pentru a scrie corect, in acest scop, sirul in pipe, trebuie apelat:
a) write(pfd[0], s, strlen(s)); b) write(pfd[0], &s, sizeof(s)); c) write(pfd[1], s, strlen(s)); d) write(pfd[1], s, sizeof(s)); e) write(pfd[1], &s, strlen(s));

10. Un router care poate fi administrat folosind un browser Web are sistem de operare?
a) da, pentru ca e un dispozitiv electronic; b) da, pentru ca e un sistem de calcul cu procesor si dispozitive periferice; c) nu, pentru ca e deservit de un firmware; d) nu, pentru ca este un dispozitiv electronic de uz general; e) nu, pentru ca e format din componente proprietare

11. Linia urmatoare:

```
echo `cat f | wc -l` > g
```

a) scrie intr-un fisier dimensiunea (in octeti) unui alt fisier; b) afiseaza pe ecran dimensiunea unui fisier; c) compara numarul de linii dintr-un fisier cu o variabila g; d) scrie intr-un fisier numarul de linii din alt fisier; e) scrie intr-un fisier sirul de caractere cat f | wc -l

12. Planificarea la executie a proceselor este
a) un apel sistem; b) o comanda care planifica executia programelor la diferite momente de timp; c) o functionalitate implementata de sistemul de operare; d) o facilitate existenta in toate interpretoarele de comenzi moderne; e) un algoritm de gestiune a drepturilor de executie pentru programe

13. Un program C primeste parametrii din linia de comanda in variabila argv. Daca e compilat si apelat ca mai jos, care e argv[2]?

```
./program "1 2" "3 1" 2 3 '1' 2 3
```

a) 1 b) 2 c) 1 2 d) 3 1 e) 3

14. Comanda `grep -E '^.[st].+\.t[empxt]*$'` primind pe cate o linie sirurile de mai jos le va afisa la iesirea standard pe urmatoarele:
a) conf.t b) s1.text c) config.tmp d) s1.ttm e) os1.text

15. Codul de mai jos va afisa la iesirea standard (alegeti varianta corecta):

```
int main(int argc, char** argv){ int pid1, pid2;
if ((pid1 = fork()) < 0) { printf("Error\n"); exit(1); }
printf("C\n");
if ((pid2 = fork()) < 0) { printf("Error\n"); exit(2); }
printf("B\n");
return 0;}
```

a) 1 caracter B si 2 caractere C; b) 4 caractere B si 2 caractere C; c) 2 caractere B si 1 caracter C; d) 1 caracter B si 1 caracter C; e) 3 caractere B si 3 caractere C

OS Exam Example, 2025/2026

Only one answer is correct for each of the multiple-choice questions below. A question with a correct and complete answer receives 10 points.
*The final grade is calculated with the formula G=2.5+r/20, where r is the number of correctly answered questions. Note that the so-called "punct din oficiu" is included: 2.5=**1**+1.5.*
Open-ended questions require answers in a few phrases, and may have variable scores. Their count varies across presentations and generally increases with the presentation number. Read carefully. Don't mix up related but distinct concepts. Example: command != function != process
Required working time: 20-25 minutes. Allotted working time: 35 minutes.

1. Explain the functional difference between the following command lines:
```
$  ls -l && tail abc.txt && test -f fis1.c
$  ls -l || tail abc.txt || test -f fis1.c
```

2. Enumerate at least 4 services offered by operating systems.

3. The Java Virtual Machine (JVM) is a virtual machine
a) of type system; b) of type process; c) container; d) implemented inside the UNIX operating system kernel; e) specific to UNIX

4. Information about file access rights is found:
a) in its associated index node; b) in the file; c) in the directory containing the file; d) in the operating system; e) in the symbolic link

5. The kill() function
a) always terminates a process abruptly; b) sends a signal; c) destroys a process without right of appeal; d) captures a signal; e) sends signal number 9 to the current command

6. The alarm() function
a) schedules the sending of a specific signal in the future; b) signals an error situation that occurred during execution; c) resets a triggered alarm; d) informs a process about the occurrence of an error in another process; e) announces processes that they must terminate immediately

7. Hard links are:
a) different directory entries that refer to the same index node; b) alternative names for the same file, always located in the same directory; c) individual files that refer to other files; d) individual files that refer to fixed directories; e) special files

8. The fork() function creates:
a) a command different from the parent program, but initially identical to it; b) a child of the current program loaded from disk; c) a new process with the same identifier as the current process; d) a new process; e) a parent process

9. A programmer wants to transmit a variable-length string smaller than 100, via a pipe. The string is contained in a variable char s[100] and was initialized by calling strcpy(), and the descriptors associated with the pipe are in the variable pfd. To correctly write the string to the pipe for this purpose, one must call:
a) write(pfd[0], s, strlen(s)); b) write(pfd[0], &s, sizeof(s)); c) write(pfd[1], s, strlen(s)); d) write(pfd[1], s, sizeof(s)); e) write(pfd[1], &s, strlen(s));

10. Does a router that can be administered using a Web browser have an operating system?
a) yes, because it is an electronic device; b) yes, because it is a computing system with a processor and peripheral devices; c) no, because it is driven by a firmware; d) no, because it is a general-purpose electronic device; e) no, because it is formed of proprietary components.

11. The following line:
```
echo `cat f | wc –l` > g
```
a) writes the size (in bytes) of another file into a file; b) displays the size of a file on the screen; c) compares the number of lines in a file with a variable g; d) writes the number of lines from another file into a file; e) writes the character string cat f | wc -l into a file

12. Process scheduling is
a) a system call; b) a command that schedules the execution of programs at different times; c) a functionality implemented by the operating system; d) a facility existing in all modern command interpreters; e) an algorithm for managing execution rights for programs

13. A C program receives command line parameters in the argv variable. If it is compiled and called as below, what is argv[2]?
```
./program "1 2" "3 1" 2 3 '1' 2 3
```
a) 1 b) 2 c) 1 2 d) 3 1 e) 3

14. The command `grep -E '^.[st].+\.t[empxt]*$'` receiving the strings below on a separate line will display the following on the standard output:
a) conf.t b) s1.text c) config.tmp d) s1.ttm e) os1.text

15. The code below will display the following on the standard output (choose the correct option):
```
int main(int argc, char** argv){ int pid1, pid2;
if ((pid1 = fork()) < 0) { printf("Error\n"); exit(1); }
printf("C\n");
if ((pid2 = fork()) < 0) { printf("Error\n"); exit(2); }
printf("B\n");
return 0;}
```
a) 1 B character and 2 C characters; b) 4 B characters and 2 C characters; c) 2 B characters and 1 C character; d) 1 B character and 1 C character; e) 3 B characters and 3 C characters